

When Should Multimodal Agents Spend More Test-Time Compute?

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Abstract

Multimodal agents can spend extra inference budget by sampling multiple actions, planning before acting, reflecting after a failed attempt, or selecting among candidate trajectories with a verifier, but the same extra budget is unlikely to help every example. This paper studies adaptive test-time compute routing for offline multimodal agent tasks. We introduce ATCoR, a training-free router that uses benchmark-visible source cues and first-pass diagnostics to choose a budget tier and a test-time operator without using gold labels. The planned evaluation covers ScreenSpot-Pro/RegionFocus GUI grounding, multimodal Mind2Web offline web action prediction, and OmniGUI offline smartphone tasks under matched-budget baselines. The full result is **TBD_HEADLINE_RESULT**: this scaffold intentionally leaves outcome-dependent claims as placeholders until terminal run artifacts and canonical analysis tables exist.

1 Introduction

Multimodal agents increasingly operate over screenshots, rendered interfaces, HTML or accessibility metadata, and natural-language instructions. In these settings, extra test-time compute can take several forms: sampling multiple candidate actions, explicitly planning before an action, retrying after reflection, or ranking trajectories with a verifier. The central question for this paper is not whether more compute can help in the abstract, but when a multimodal agent should spend that compute on a particular task instance.

Uniform scaling is a weak default for this question. Self-consistency and deliberate trajectory search show that multiple samples can improve reasoning when independent samples expose useful variation (Wang et al., 2022; Yao et al., 2023). Reasoning-and-acting and reflection loops give agents mechanisms for recovery (Yao et al., 2022; Shinn et al., 2023). At the same time, GUI and web-agent benchmarks expose visually grounded errors,

action-space ambiguity, and missing-evidence failures that may not be recoverable by asking the same model to think longer (Luo et al., 2025; Deng et al., 2023; Henry et al., 2026). This makes adaptive allocation a measurement problem: the controller must distinguish recoverable uncertainty from cases where additional calls only add cost or amplify verifier mistakes.

We study this problem with ATCoR, a training-free adaptive test-time compute router. ATCoR observes benchmark-visible cues such as modality mix, interface density, instruction specificity, and expected action horizon, then runs a cheap first-pass probe to measure parse validity, confidence, trajectory diversity, and verifier agreement. It maps those signals to a budget tier and one of several operators: direct action, self-consistency, plan-then-act, act-reflect-retry, or verifier-based selection. The router is designed to be auditable: every row should record the selected operator, budget accounting, prompt serialization, parser status, and verifier trace.

The paper-facing evaluation is still in the run stage. The selected benchmark families are ScreenSpot-Pro/RegionFocus, multimodal Mind2Web, and OmniGUI offline smartphone tasks, replacing an earlier InterLV plan according to the run-stage matrix. The headline comparison, per-family deltas, and statistical tests are **TBD_HEADLINE_RESULT** and **TBD_SIGNIFICANCE**; they will be backfilled only after the required terminal raw rows and analysis artifacts exist. Our intended contributions are: (1) a concrete adaptive routing protocol for multimodal agent test-time compute, (2) a matched-budget evaluation design across offline multimodal benchmarks, and (3) an analysis of **TBD_HARMFUL_SCALING** cases where extra compute fails or hurts.

2 Related Work

Test-time compute scaling. Chain-of-thought prompting, self-consistency, and tree-style reasoning search motivate the use of extra samples or reasoning paths at inference time (Wei et al., 2022; Wang et al., 2022; Yao et al., 2023). Recent adaptive-inference work studies budget-quality tradeoffs and dynamic thinking budgets (Nogueira et al., 2025; Xu et al., 2026, 2025; Guo et al., 2026). ATCoR differs in scope: it routes among agent operators on multimodal tasks rather than allocating a single textual reasoning depth.

Agent loops and recovery operators. ReAct-style loops connect reasoning to actions, while reflection methods allow a model to revise behavior after a failed attempt (Yao et al., 2022; Shinn et al., 2023). Verifier-based selection and self-verification are natural candidates for spending additional calls on agentic tasks (Zhao et al., 2026). The open question for this paper is whether a cheap difficulty estimate can decide which operator is worth its cost on each example.

Multimodal GUI and web benchmarks. GUI and web-agent benchmarks create a setting where errors may arise from visual grounding, action parsing, long-horizon planning, or interface-specific ambiguity. Mind2Web and WebArena anchor web-agent evaluation (Deng et al., 2023; Zhou et al., 2023), while RegionFocus and OmniGUI motivate visual and smartphone-focused offline settings (Luo et al., 2025; Henry et al., 2026). The present study uses these families to test not only success rate, but whether extra compute is spent on examples where it can plausibly change the outcome.

3 Method: Adaptive Test-Time Compute Routing

ATCoR is a training-free controller that maps each task instance to a budget tier and a test-time operator. The controller uses two classes of signals. Source cues are available before model inference: benchmark family, modality mix, screenshot or UI complexity, instruction specificity, candidate-action density, and expected horizon. First-pass diagnostics are produced by a cheap probe: parse validity, direct-answer confidence, diversity across sampled candidates when available, and verifier agreement. The controller must not inspect gold labels, hidden answers, or post-hoc success outcomes.

Given these signals, ATCoR chooses among five operators. Direct action spends one call and is used when the probe is parse-valid and confident. Self-consistency samples several candidate actions and aggregates them. Plan-then-act asks the model to derive a short plan before outputting the final action. Act-reflect-retry lets the model critique and revise an initial attempt. Verifier-select generates candidate trajectories and chooses one using a verifier. The intended Figure 1 is **PLACEHOLDER_FIGURE1_IMAGE2**; it must be generated through the image-2 pipeline before final drafting.

4 Experimental Setup

The planned evaluation uses the manager-certified selected families in experiments/MATRIX.json: ScreenSpot-Pro/RegionFocus, multimodal Mind2Web, and OmniGUI offline smartphone tasks. The earlier InterLV family is not treated as a selected run-stage family in this scaffold. Each benchmark condition should compare ATCoR against matched-slice baselines: single pass, uniform self-consistency, uniform plan-then-act, uniform reflect/retry, uniform verifier-select, a random-budget diagnostic router, and the planned ablations. The final task counts are **TBD_TASK_COUNTS**.

The primary metric is matched-budget task success. Secondary metrics include area under the budget-accuracy curve, cost per successful example, wasted-extra-compute rate, harmful-scaling rate, verifier calibration, and paired uncertainty estimates. Every result must be generated from terminal raw rows, not from pilot status counts. At the time of this scaffold, the repaired ScreenSpot-Pro ATCoR pilot is still live; its current rows are diagnostic run-stage evidence, not final paper-facing results.

5 Main Results

The main result will report whether ATCoR improves matched-budget success over the strongest feasible uniform baseline. The pooled headline result is **TBD_HEADLINE_RESULT**; paired uncertainty is **TBD_SIGNIFICANCE**. Per-family outcomes are **TBD_SCREENSPOT_DELTA**, **TBD_MIND2WEB_DELTA**, and **TBD_OMNIGUI_DELTA**. Until all required selected-family conditions are terminal and collected, this section must remain a placeholder

PLACEHOLDER_FIGURE1_IMAGE2: image-2 method overview to be generated with full provenance.

Figure 1: PLACEHOLDER_FIGURE1_IMAGE2. ATCoR routes each multimodal task to a budget tier and operator using source cues plus a first-pass diagnostic probe.

Family	Tasks	Methods	Backend	Budget	Metric
ScreenSpot-Pro / RegionFocus	TBD	ATCoR + baselines	TBD	matched calls	TBD
Mind2Web	TBD	ATCoR + baselines	TBD	matched calls	TBD
OmniGUI	TBD	ATCoR + baselines	TBD	matched calls	TBD

Table 1: Planned cross-benchmark evidence matrix. Numerical entries are placeholders until terminal raw rows and canonical analysis artifacts exist; exact placeholder owners are listed in paper/RESULT_PLACEHOLDERS.md.

rather than a positive-result narrative.

6 Analysis and Ablations

The analysis will characterize where extra compute helps and where it fails. The planned budget curve is **TBD_BUDGET_CURVE**. The planned verifier-calibration result is **TBD_VERIFIER_CALIBRATION**. The harmful-scaling and wasted-extra-compute findings are **TBD_HARMFUL_SCALING**. Ablations will remove source cues, first-pass diagnostics, operator choice, and abstention when the corresponding run cells are available.

7 Failure Cases

Failure cases will be selected only from saved terminal rows. Candidate categories include visual ambiguity, parser instability, verifier overconfidence, unrecoverable missing evidence, and budget spent on examples whose first-pass error is not recoverable by the available operators. The qualitative examples are **PLACEHOLDER_CASES**.

8 Conclusion

This scaffold sets up an EMNLP/ACL paper about adaptive test-time compute for multimodal agents, but it does not yet claim that ATCoR improves accuracy. The final conclusion will be written only after **TBD_HEADLINE_RESULT**, **TBD_SIGNIFICANCE**, and the failure analysis are backed by terminal full-matrix evidence.

Limitations

This draft is intentionally limited by run-stage evidence availability. The selected benchmark families and metrics are defined, but full-scale terminal rows are not yet complete. Any final claim about gains, significance, calibration, or failure rates is therefore blocked until the run-stage evidence bundle and analysis tables are refreshed.

Ethical Considerations

The planned experiments use offline benchmark tasks and saved media or metadata rather than live user interaction. The manuscript should not expose private API routes, local device IDs, cache paths, or daemon supervision details. Any benchmark-specific licensing or redistribution limits must be preserved in the final reproducibility appendix.

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A Reproducibility Appendix

The final appendix will list benchmark provenance, task filtering, run manifests, prompt templates, model/backend settings in paper-facing terms, collection commands, and STOP-file behavior. Current run-stage collection status is **TBD_TASK_COUNTS**; terminal collection artifacts are required before this appendix can be complete.